Momentum

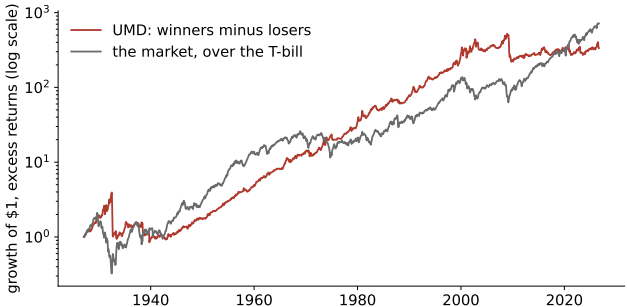
Buy what went up

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NYU Stern — Data-Driven Investing

Fall 2026

Buy what went up



- Every month: buy last year's winners, short last year's losers
- \$1 in 1927 grows to \$335. The market: \$719
- Same Sharpe ratio as the market: 0.45

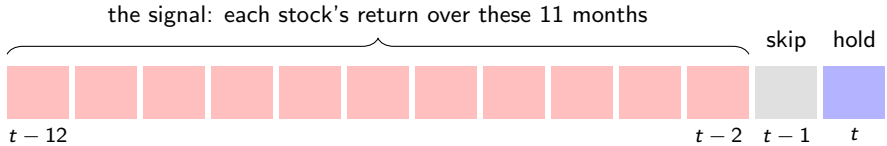
Should this work?

A No: everyone can see it

B Yes: it is risky

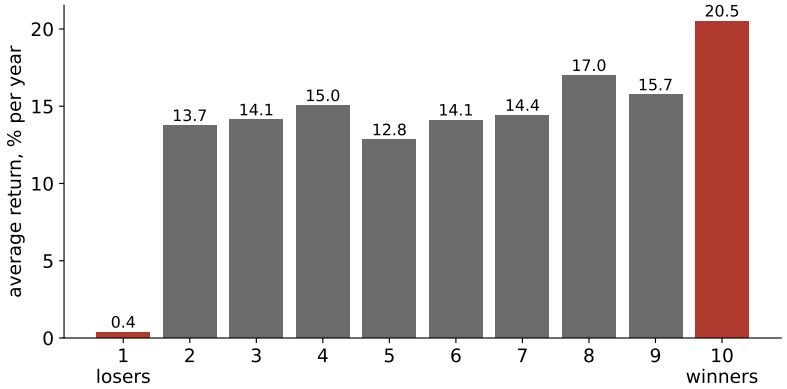
C Yes: people are slow

The rule



- Rank every stock on the signal. Buy the top 10%, short the bottom 10%.
- Do it again next month
- Why skip last month? Last month's winners tend to reverse. Hold that thought.

Winners beat losers, and the losers do most of the work



- Top minus bottom decile: 20% a year, Sharpe ratio 1.03 (our panel, 1981–2000, value-weighted)
- The middle deciles earn about 13.5%. The losers earn 0.4%.

Adjusting for risk makes it bigger

$$UMD_t = \alpha + \beta \left(R_t^m - r_t^f\right) + \varepsilon_t$$

UMD_t: winners minus losers in month t $\quad$ $R_t^m - r_t^f$: the market over the T-bill

UMD, 1927–2026	average return	α	β
per year	7.4%	9.9% (t = 6.4)	−0.30

- The losers are the stocks that fell with the market. Short them and you are short the market.

Momentum's alpha is larger than its average return

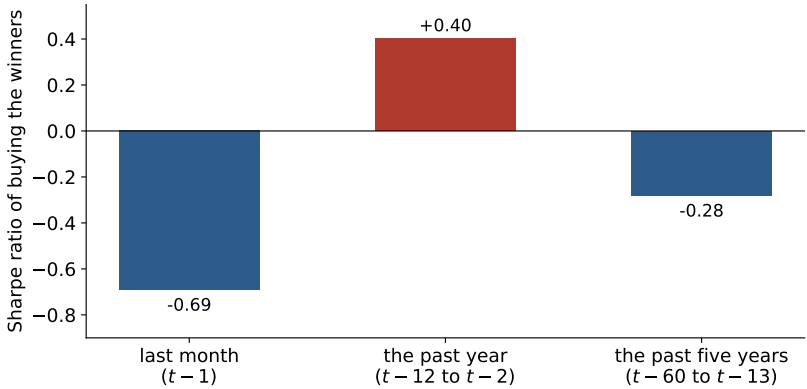
The fourth factor

- Carhart (1997) added UMD to the market, size and value
- Since then, a new strategy has to beat momentum too:

$$r_t = \alpha + \beta_m \left(R_t^m - r_t^f\right) + \beta_u \, UMD_t + \varepsilon_t$$

- β_u near 1 and α near 0: the strategy is momentum under another name
- Today we ask it of every version: is it anything more than UMD?

Horizon matters



- Last month reverses. The past year continues. The past five years reverse.
- That is why the signal skips $t-1$ (stocks, 1931–2026, Fama–French factors)

Is it risk?

- A risk premium is pay for losing money in bad times
- Momentum does the opposite. In the worst third of market months UMD earns $+1.3\%$ a month; in the best third, -0.6%
- And the winners have just gone up: if anything they got safer, not riskier

So the simple risk story does not work

Or is it behavior?

- **Underreaction**: news gets into prices slowly
- **The disposition effect**: investors sell winners too soon and hold on to losers, so prices adjust in steps (Frazzini 2006)
- **Stronger where news travels slowly**: small stocks, few analysts (Hong, Lim and Stein 2000)

Then why has no one traded it away? Because it crashes. Hold that thought too.

It survives out of sample

UMD		average	Sharpe	CAPM α
1927–1964	before the paper's data	7.7%	0.41	11.9% (t = 4.6)
1965–1989	the paper's sample	9.9%	0.80	9.8% (t = 4.0)
1990–2026	after publication	5.3%	0.33	7.9% (t = 3.0)

- Weaker since publication, still there
- And in other markets: equity indices, currencies, bonds, commodities. All asset classes combined: a Sharpe ratio of 0.81, 1972–2010 (Asness, Moskowitz and Pedersen 2013)

So everyone built a version

variation	what changes	paper
cross-sectional	the original: rank on $t-12$ to $t-2$	Jegadeesh and Titman 1993
earnings momentum	rank on the announcement return	Chan, Jegadeesh, Lakonishok 1996
industry momentum	rank industries, not stocks	Moskowitz and Grinblatt 1999
slow diffusion	stronger where coverage is thin	Hong, Lim and Stein 2000
52-week high	distance from the 52-week high	George and Hwang 2004
residual momentum	momentum in model residuals	Blitz, Huij and Martens 2011
the echo	months 7–12 only	Novy-Marx 2012
time-series momentum	your own past return, not a rank	Moskowitz, Ooi, Pedersen 2012
everywhere	bonds, currencies, commodities	Asness, Moskowitz, Pedersen 2013
risk-managed	scale by trailing volatility	Barroso and Santa-Clara 2015

The full list

Many papers. How many trades?

1980–2000	Sharpe	corr. with 12-month momentum	α on market + UMD
6-month momentum	0.61	0.79	-0.2% $(t = -0.1)$
residual momentum	0.64	0.53	2.5% $(t = 1.0)$
earnings momentum	1.17	0.24	10.1% $(t = 4.5)$

- Six-month momentum is UMD again. Earnings momentum is a different trade.
- You cannot tell from the name. Run the regression.

Industries

Moskowitz and Grinblatt (1999): *Do industries explain momentum?*

- Industries have momentum too: buy last year's best industries, short the worst
- Their claim: much of stock momentum is industry momentum
- If so, momentum is a bet on a few sectors, and less diversified than it looks

Our question: industry momentum since 1927. Is it anything more than UMD?

Now: the notebook

Notebook: Momentum — one strategy, three ways, the same two regressions

1. **Stock momentum.** Four steps on our panel, then the CAPM and the market + UMD regressions.
2. **Industry momentum, 1927–2026.** Build it with AI, check it, then the same two regressions.

Bring back: is industry momentum a different trade from stock momentum?

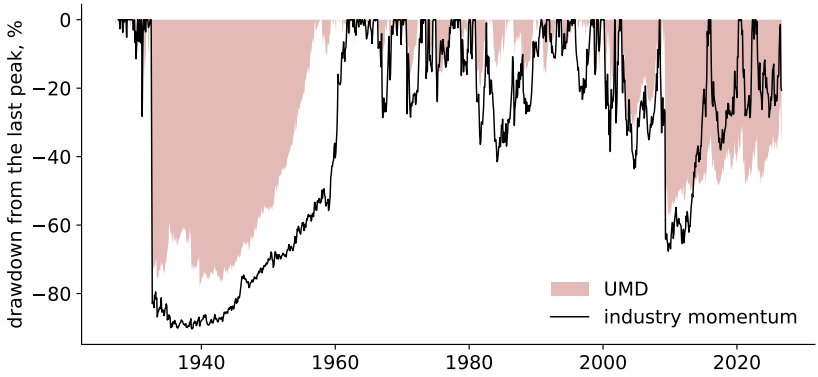
Stop at **Back to the slides**; the challenge after it is homework.

Industry momentum is mostly UMD

	CAPM α	α on market + UMD	UMD β
stock momentum, 1981–2000	19.0% (t = 4.3)	5.1% (t = 2.7)	1.45
industry momentum, 1927–2026	13.2% (t = 5.9)	2.9% (t = 1.8)	1.05

- Over a century, the industry bet and the stock bet are mostly one trade (correlation 0.74)
- Our stock deciles load more than 1 on UMD because they are more extreme than UMD's 30/70 split

Both crash, at the same time



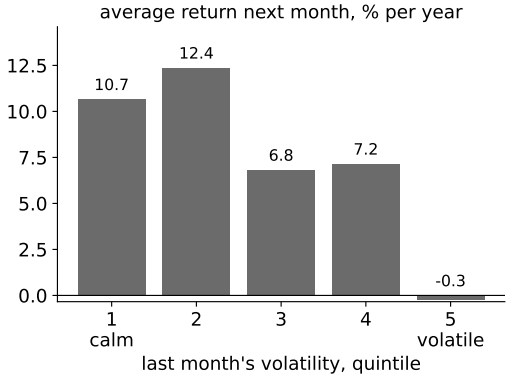
- 1932: UMD -65%, industry momentum -71%. The market: -9%.
- 2009: UMD -53%, industry momentum -50%. The market: $+29\%$.

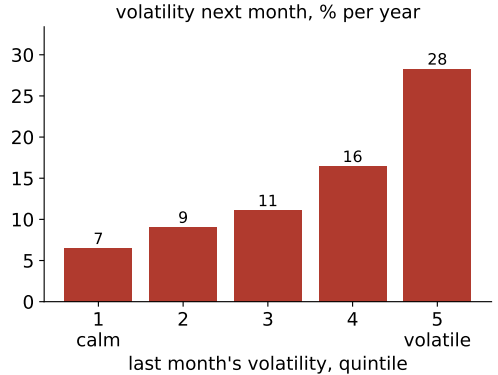
Why it crashes

UMD, 1927–2026	months	market β	average return
after a two-year bear market	191	-0.68	-9.3% a year
after a two-year bull market	1,005	0.01	$+10.5\%$ a year

- After a crash, the losers are the stocks that fell the most: high-beta stocks
- Short them, and you are short the rebound. In the months the market rose after a bear market, UMD lost 4.1% a month
- Momentum is short a call on the market's recovery (Daniel and Moskowitz 2016)

After a volatile month: more risk, no more return





- Volatility persists: this month's variance tracks last month's (correlation 0.78, in logs)
- The average return does not follow. Sharpe ratio: 1.6 after the calmest months, 0 after the most volatile

So hold less after a volatile month

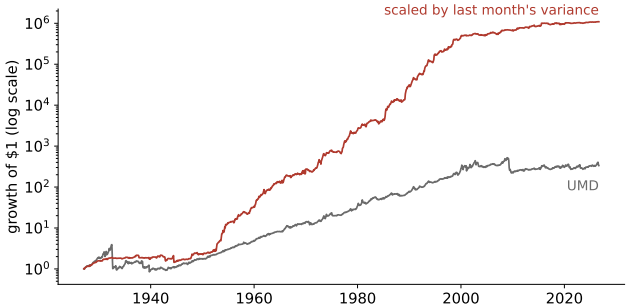
$$UMD_t^{\sigma} = \frac{c}{RV_{t-1}} UMD_t$$

- RV_{t-1}: the variance of momentum's daily returns last month
- c: a constant, set so UMD^{σ} has the same volatility as UMD. It sets the scale, not the Sharpe ratio
- Barroso and Santa-Clara (2015); Moreira and Muir (2017)

You do not need to forecast returns. Only volatility, and volatility is predictable.

Details

Twice the Sharpe ratio, half the drawdown



Same volatility as UMD, 1927–2026

- Sharpe ratio: 0.45 $\rightarrow$ 0.94
- Worst drawdown: $-78\% \rightarrow -33\%$
- 1932: -0.5%. 2009: -0.2%.
- α on market + UMD: 10.4% a year (t = 7.1)

The catch

- **Leverage.** The position is above $2\times$ in a quarter of months, and above $8\times$ in the calmest 1%
- **Trading.** The position moves by 0.9 a month on average
- **Hindsight.** c uses the whole sample. It does not change the Sharpe ratio, but it does set the leverage

Cap the position at $2\times$: Sharpe ratio 0.97, worst drawdown -20%

What to take away

- Buy the past year's winners, skip last month: a century of returns, with a negative beta
- Not a risk premium in the usual sense. Most likely, slow reaction to news
- Many versions, fewer trades: regress on UMD before you believe a new one
- It crashes in rebounds, when volatility is high. Scale by last month's variance.

Homework: the challenge at the end of the notebook. Volatility-managed industry momentum.

Appendix

A1: The many momenta, the longer list

variation	what changes	paper
cross-sectional (12,1)	the original	Jegadeesh and Titman 1993, *JF*
UMD as a factor	a benchmark, not a trade	Carhart 1997, *JF*
earnings momentum	the signal is the announcement return, not the price	Chan, Jegadeesh and Lakonishok 1996, *JF*
industry momentum	rank industries, not stocks	Moskowitz and Grinblatt 1999, *JF*
slow information diffusion	stronger where coverage is thin	Hong, Lim and Stein 2000, *JF*
volume-conditioned	interact the signal with turnover	Lee and Swaminathan 2000, *JF*
beta-hedged	strip out the time-varying market exposure	Grundy and Martin 2001, *RFS*
52-week high	distance from a high, not a cumulative return	George and Hwang 2004, *JF*
residual momentum	momentum in Fama–French residuals	Blitz, Huij and Martens 2011, *JEmpFin*
the echo	months 7–12 only, not 2–12	Novy-Marx 2012, *JFE*
time-series momentum	your own past return, not a relative rank	Moskowitz, Ooi and Pedersen 2012, *JFE*
everywhere	the same signal in bonds, FX, commodities	Asness, Moskowitz and Pedersen 2013, *JF*
frog in the pan	gradual news beats sudden news	Da, Gurun and Warachka 2014, *RFS*
risk-managed	scale the position by trailing volatility	Barroso and Santa-Clara 2015, *JFE*
crashes	when and why it fails	Daniel and Moskowitz 2016, *JFE*

Back

A2: How the volatility-managed version is built

- Daily UMD returns from Ken French's library; for each month, $RV_t = \sum_{d \in t} (r_d - \bar{r}_t)^2$
- Weight for month t: c/RV_{t-1}, known at the start of the month
- c makes the managed series as volatile as UMD over 1927–2026
- Weight on UMD: median 0.84, 90th percentile 3.7, 99th percentile 8.1
- CAPM α: 15.4% a year (t = 9.4). On market + UMD: 10.4% (t = 7.1), UMD β 0.51

Back

A3: Data

- **Factors**: market, T-bill, UMD, short- and long-term reversal; monthly and daily, July 1926 to August 2026. Ken French's data library.
- **Industries**: the 49 value-weighted industry portfolios, monthly and daily, from the same library. Industry momentum: rank on $t-12$ to $t-2$, the top 5 minus the bottom 5, equal-weighted.
- **Stocks**: our CRSP panel, 1980–2000. Common stocks on NYSE, AMEX and Nasdaq; deciles on NYSE breakpoints, value-weighted.
- **The momentum variants**: the published long-shorts from the factor-zoo lecture, 1980–2000.